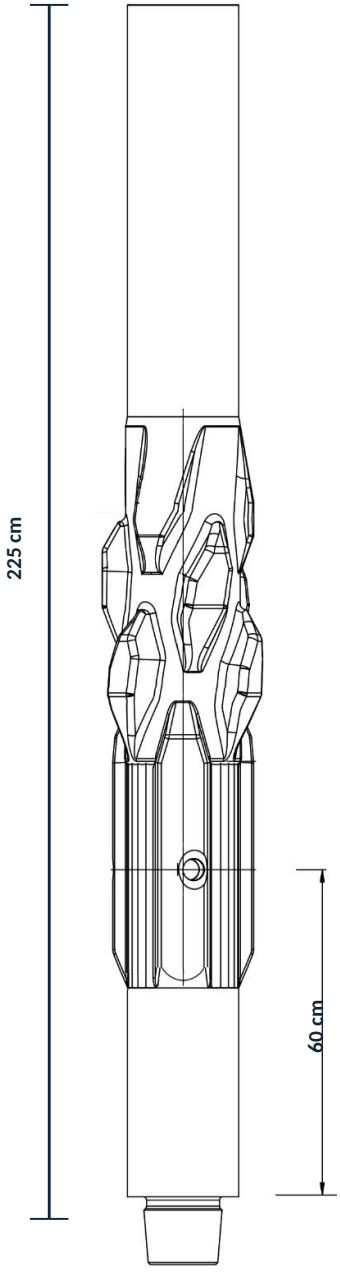
Casing Integrity | Wellbore Imaging | Cement Evaluation | Hole Cleaning

ECCO® ENP3 imaging tool with hydraulic enhancement

Non-magnetic ultrasonic imaging with integrated hole-cleaning module

Overview

The ECCO® ENP3 from MIT Technologies verifies open-hole geometry, casing condition and cement integrity while eliminating additional wireline runs. It extends the ECCO platform with a non-magnetic body and a proprietary three-stage hydraulic enhancement module that improves borehole cleaning and reduces pack-off risk.



ENP3 is a Memory While Drilling system based on multi-frequency, multi-sensor ultrasonic pulse-echo technology. Three high-frequency imaging sensors, spaced 120° apart on a single sensor plane located 60 cm above the lower (pin) shoulder, record borehole geometry in open hole and map cement behind casing in cased hole.

A dedicated mud sensor performs in-situ logging of mud acoustic properties, providing real-time velocity correction of every imaging measurement.

The tool records independently of LWD/MWD in the BHA, in both rotating and sliding modes, with optimized dynamic resolution at drilling speeds up to 180 RPM and selective command during tripping without rotation.

The hydraulic module — accelerator, twisted-pad and diffuser stages — conditions the annular flow around the tool, keeping cuttings in suspension and eliminating the low-pressure wake zone where pack-off typically initiates.

Left: ENP3 general arrangement (not to scale). Overall length 225 cm; imaging sensor plane 60 cm from pin shoulder.

BENEFITS

- 4D continuous ultrasonic open-hole geometry mapping.
- Eliminates expensive wireline, LWD & caliper runs; reduces rig time and operational exposure.
- Cement evaluation, casing inspection and borehole imaging in one run.
- Identifies Top of Cement (TOC) and cement bond behind casing with 360° imaging, plus casing condition assessment.
- Records in both rotating and sliding modes.
- In-situ mud property logging and acoustic velocity correction.
- Improves hole cleaning under rotation, tripping or circulation — each acting independently.
- Reduces pack-off risk by eliminating the low-pressure wake zone.
- Non-magnetic body for BHA placement near directional sensors.
- Improves liner hanger placement confidence in highly deviated wells.

Hydraulic Enhancement Module

The ENP3 proprietary hydraulic module is built as three sequential stages that optimize the annular mud pressure and flow profile around the tool, improving cuttings transport and reducing pack-off risk.

STAGE 1

Hydraulic accelerators

Three pads accelerate the annular flow entering the module, raising local velocity to lift and transport cuttings.

STAGE 2

Twisted pads

Two sets of twisted pads generate optimized mud jetting and controlled turbulence, preventing cuttings beds from settling.

STAGE 3

Diffuser

Recovers flow pressure and eliminates the low-pressure wake zone behind the pads — the primary initiation point for pack-off.

Independent hole-cleaning action

Each operating condition activates the module on its own — hole cleaning does not depend on rotation and circulation occurring together.

Operation	Hole-cleaning mechanism
Rotation only	Twisted pads act as an impeller, imparting tangential and axial momentum to the mud and agitating cuttings beds on the low side.
Tripping only	Axial movement drives mud across the twisted pads, which act as a turbine and educt the surrounding mud — no rotation or circulation required.
Circulation only	Annular flow is accelerated, jetted and diffused through the three stages with no string rotation, maintaining cuttings transport while sliding.
Combined	Effects are additive under any combination of rotation, tripping and circulation.

Imaging & sensing

- 3 ultrasonic pulse-echo imaging sensors at 120° on one sensor stage
- Sensor plane 60 cm from lower (pin) shoulder
- Integrated mud sensor for in-situ mud property logging
- Multi-sensor string eccentricity compensation
- Application-specific multi-frequency tuning

Applications

- Open-hole geometry: enlargement, reduction and asymmetry
- Cement mapping and TOC behind casing
- Casing inspection
- Pre-liner hole assessment in highly deviated wells
- Recording in rotating and sliding modes

ECCO family

Model	Material	Body / pad configuration
ENP1	Alloy	Open-wrap 30° twisted blades, 4 pads
ENP2	Non-mag	Slick body
ENP3	Non-mag	Three-stage hydraulic enhancement module + mud sensor

SPECIFICATION SHEET

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Dimensions and Specifications

Dimensions	Model	**ENP 3**
	Material	Non Mag
	Maximum OD	12 in (at central pads)
	Fishing Neck OD	8.25 in
	Minimum ID	2.25 in
	Overall length [1]	2.25 m
Ratings	Over pull (max)	1,000,000 lbs
	Tensile load (max)	1,000,000 lbs
	Tensile torque	200,000 ft-lbs
	Pressure Hydrostatic (max)	20,000 psi
	Temperature (max) [2]	150 Deg C / 300 Deg F
	Vibration	20 G
	Minimum Casing Internal Diameter	12.3 in
	Borehole Fluid Density	Up to 14 ppg
	Borehole Fluid Type	WBM/OBM/SBM
Sensors	Acquisition modes	Rotating and sliding
	Imaging sensors	3 × ultrasonic pulse-echo, 120° apart, one stage
	Sensor plane position	60 cm from lower (pin) shoulder
	Mud sensor	In-situ mud property logging
	Azimuthal sampling	108 bins per revolution (360° imaging)
Connection	ECCO Top Conn.	6-5/8” F.H. Box
	ECCO Bottom Conn.	6-5/8” F.H. Pin
	Make-up Torque [3]	37,000 ft-lbs
	Cross-over [4]	6-5/8” Regular

[1] Actual tool length could be slightly different.
[2] Max. temperature is bound by battery rating up to 150 degC max.
[3] Make-up torque for new tool. Torque to be reduced based on connection wear as per API guideline.
[4] Cross-over can be provided upon request. Cross-over to any thread type can be arranged, providing sufficient notice.

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